

COMP 3311

DATABASE MANAGEMENT

SYSTEMS

LECTURE 8 EXERCISES

STRUCTURED QUERY LANGUAGE (SQL):

DATA MANIPULATION LANGUAGE (DML)

BOATRESERVATIONS RELATIONAL DATABASE

Sailor(sailorId, sName, hkid, rating, age)

Reserves(sailorId, boatId, rDate)

Boat(boatId, bName, color, bType)

Attribute names in italics are foreign key attributes.

Sailor

<u>sailorId</u>	sName	hkid	rating	age
10	Emily	P936219(3)	null	16
15	Zoey	K091876(5)	null	17
22	Dustin	A298456(6)	7	45
29	Brutus	D746239(0)	1	33
31	Lucy	P930281(5)	10	55
32	Andy	M028391(4)	8	25
58	Rachael	A819204(3)	10	17
64	Horatio	F710943(2)	7	35
71	Zorba	P910562(4)	1	16
74	Horatio	T810463(1)	9	35
85	Amy	B719037(8)	3	25
95	Bob	E012735(5)	3	63
99	Chris	F710943(2)	10	30

13 tuples

Reserves

<u>sailorId</u>	<u>boatId</u>	<u>rDate</u>
22	103	08-SEP-25
22	104	10-OCT-25
22	101	10-OCT-25
22	102	10-NOV-25
31	102	10-OCT-25
31	103	06-NOV-25
31	102	12-NOV-25
64	101	05-SEP-25
64	102	08-SEP-25
74	103	08-SEP-25
99	102	08-AUG-25

11 tuples

Boat

<u>boatId</u>	bName	color	bType
101	Interlake	blue	dinghy
102	Interlake	red	sloop
103	Clipper	green	dinghy
104	Marine	red	catamaran
105	Serenity	cyan	null

5 tuples

EXERCISE 1

Find the sailors who have a rating of 8 or greater and are 30 years old or younger. Include in the query result tuples attributes sailor name, age and rating.

☞ (Andy, 25, 8) (Rachael, 17, 10) (Chris, 30, 10)

```
select sName, age, rating
from Sailor
where rating >= 8 and age <= 30
```

<u>sailorId</u>	sName	hkid	age	rating
32	Andy	M028391(4)	25	8
58	Rachael	A819204(3)	17	10
99	Chris	H814243(7)	30	10

Get only those tuples where the rating ≥ 8 and the age ≤ 30 .

Query result

sName	age	rating
Andy	25	8
Rachael	17	10
Chris	30	10

Project on sName, rating and age.

EXERCISE 2

Find the sailors who have reserved boat 103.
Include in the query result tuples attribute sailor name.

☞ (Dustin) (Lucy) (Horatio)

How to eliminate
duplicate columns
in the join result?

Join Sailor and
Reserves on sailorId.

```
select sName
from Sailor, Reserves
where Sailor.sailorId=Reserves.sailorId
and boatId=103
```

Query
result

sName
Dustin
Lucy
Horatio

Project on
sName.

sailorId	sName	hkid	rating	age	sailorId1	boatId	rDate
22	Dustin	A298456(6)	7	45	22	103	08-SEP-25
22	Dustin	A298456(6)	7	45	22	104	10-OCT-25
22	Dustin	A298456(6)	7	45	22	101	10-OCT-25
22	Dustin	A298456(6)	7	45	22	102	10-NOV-25
31	Lucy	P930281(5)	10	55	31	102	10-OCT-25
31	Lucy	P930281(5)	10	55	31	103	06-NOV-25
31	Lucy	P930281(5)	10	55	31	102	12-NOV-25
64	Horatio	F710943(2)	7	35	64	101	05-SEP-25
64	Horatio	F710943(2)	7	35	64	102	08-SEP-25
74	Horatio	T810463(1)	9	35	74	103	08-SEP-25
99	Chris	H814243(7)	10	30	99	102	08-AUG-25

sailorId	sName	hkid	rating	age	sailorId1	boatId	rDate
22	Dustin	A298456(6)	7	45	22	103	08-SEP-25
31	Lucy	P930281(5)	10	55	31	103	06-NOV-25
74	Horatio	T810463(1)	9	35	74	103	08-SEP-25

Retain only those tuples where the boatId is 103.



EXERCISE 2 (CONT'D)

Find the sailors who have reserved boat 103.
Include in the query result tuples attribute sailor name.

☞ (Dustin) (Lucy) (Horatio)

Use natural join to eliminate duplicate columns in the join result.

```
select sName
from Sailor natural join Reserves
where boatId=103
```

Join Sailor and Reserves on sailorId.

sailorId	sName	hkid	rating	age	boatId	rDate
22	Dustin	A298456(6)	7	45	103	08-SEP-25
22	Dustin	A298456(6)	7	45	104	10-OCT-25
22	Dustin	A298456(6)	7	45	101	10-OCT-25
22	Dustin	A298456(6)	7	45	102	10-NOV-25
31	Lucy	P930281(5)	10	55	102	10-OCT-25
31	Lucy	P930281(5)	10	55	103	06-NOV-25
31	Lucy	P930281(5)	10	55	102	12-NOV-25
64	Horatio	F710943(2)	7	35	101	05-SEP-25
64	Horatio	F710943(2)	7	35	102	08-SEP-25
74	Horatio	T810463(1)	9	35	103	08-SEP-25
99	Chris	H814243(7)	10	30	102	08-AUG-25

Query result

sName
Dustin
Lucy
Horatio

Project on sName.

sailorId	sName	hkid	rating	age	boatId	rDate
22	Dustin	A298456(6)	7	45	103	08-SEP-25
31	Lucy	P930281(5)	10	55	103	06-NOV-25
74	Horatio	T810463(1)	9	35	103	08-SEP-25

Retain only those tuples where the boatId is 103.



EXERCISE 3

Find the sailors who have reserved either boat 101 or boat 103.
Include in the query result tuples attributes sailor id and name.

☞ (22, Dustin) (31, Lucy) (64, Horatio) (74, Horatio)

Sailor(sailorId, sName, hkid, rating, age)

Reserves(sailorId, boatId, rDate)

Boat(boatId, bName, color, bType)

```
select distinct Sailor.sailorId, sName
from Sailor, Reserves
where Sailor.sailorId=Reserves.sailorId
and (boatId=101 or boatId=103);
```

Join Sailor
and Reserves
on sailorId.

sailorId	sName	hkid	rating	age	sailorId1	boatId	rDate
22	Dustin	A298456(6)	7	45	22	103	08-SEP-25
22	Dustin	A298456(6)	7	45	22	104	10-OCT-25
22	Dustin	A298456(6)	7	45	22	101	10-OCT-25
22	Dustin	A298456(6)	7	45	22	102	10-NOV-25
31	Lucy	P930281(5)	10	55	31	102	10-OCT-25
31	Lucy	P930281(5)	10	55	31	103	06-NOV-25
31	Lucy	P930281(5)	10	55	31	102	12-NOV-25
64	Horatio	F710943(2)	7	35	64	101	05-SEP-25
64	Horatio	F710943(2)	7	35	64	102	08-SEP-25
74	Horatio	T810463(1)	9	35	74	103	08-SEP-25
99	Chris	H814243(7)	10	30	99	102	08-AUG-25

Retain only
those tuples
where the
boat id is
101 or 103.

EXERCISE 3 (CONT'D)

Find the sailors who have reserved either boat 101 or boat 103.
Include in the query result tuples attributes sailor id and name.

☞ (22, Dustin) (31, Lucy) (64, Horatio) (74, Horatio)

Sailor(sailorId, sName, hkid, rating, age)

Reserves(sailorId, boatId, rDate)

Boat(boatId, bName, color, bType)

```
select distinct Sailor.sailorId, sName
from Sailor, Reserves
where Sailor.sailorId=Reserves.sailorId
and (boatId=101 or boatId=103);
```

sailorId	sName	hkid	rating	age	sailorId1	boatId	rDate
22	Dustin	A298456(6)	7	45	22	103	08-SEP-25
22	Dustin	A298456(6)	7	45	22	101	10-OCT-25
31	Lucy	P930281(5)	10	55	31	103	06-NOV-25
64	Horatio	F710943(2)	7	35	64	101	05-SEP-25
74	Horatio	T810463(1)	9	35	74	103	08-SEP-25

Join of Sailor and Reserves where the boat id is 101 or 103.

sailorId	sName
22	Dustin
22	Dustin
31	Lucy
64	Horatio
74	Horatio

Project on sailorId and sName.

sailorId	sName
22	Dustin
31	Lucy
64	Horatio
74	Horatio

Retain only distinct tuples.

EXERCISE 3 (CONT'D)

Find the sailors who have reserved either boat 101 or boat 103.
Include in the query result tuples attributes sailor id and name.

☞ (22, Dustin) (31, Lucy) (64, Horatio) (74, Horatio)

Sailor(sailorId, sName, hkid, rating, age)

Reserves(sailorId, boatId, rDate)

Boat(boatId, bName, color, bType)

```
select distinct Sailor.sailorId sName
from Sailor, Reserves
where Sailor.sailorId=Reserves.sailorId
and (boatId=101 or boatId=103);
```

Query result: ()

Why is it necessary to qualify sailorId in the **select** clause?

☞ **sailorId is ambiguous in the join result.**
Should we take it from Sailor or Reserves?
(For some operations it makes a difference!)

What do we get if we replace **or** with **and** in the query?

☞ **No result since there is no Boat tuple whose boatId attribute has a value that is both 101 and 103!**

How can we avoid having to qualify sailorId in the **select** clause?

☞ **Use natural join.**

EXERCISE 3 (CONT'D)

Find the sailors who have reserved either boat 101 or boat 103.
Include in the query result tuples attributes sailor id and name.

☞ (22, Dustin) (31, Lucy) (64, Horatio) (74, Horatio)

Sailor(sailorId, sName, hkid, rating, age)

Reserves(sailorId, boatId, rDate)

Boat(boatId, bName, color, bType)

```
select distinct sailorId, sName
from Sailor natural join Reserves
where boatId=101 or boatId=103;
```

Use natural join
to eliminate
duplicate columns
in the join result.

sailorId	sName	hkid	rating	age	boatId	rDate
22	Dustin	A298456(6)	7	45	103	08-SEP-25
22	Dustin	A298456(6)	7	45	101	10-OCT-25
31	Lucy	P930281(5)	10	55	103	06-NOV-25
64	Horatio	F710943(2)	7	35	101	05-SEP-25
74	Horatio	T810463(1)	9	35	103	08-SEP-25

Get only those tuples where the boat id is 101 or 103.

sailorId	sName
22	Dustin
22	Dustin
31	Lucy
64	Horatio
74	Horatio

Project on sailorId
and sName.

Query result	
sailorId	sName
22	Dustin
31	Lucy
64	Horatio
74	Horatio

Retain only
distinct tuples.

EXERCISE 4A)

Use
intersect

Find the sailors who have reserved both boat 101 and boat 103.
Include in the query result tuples attribute sailor name.

 (Dustin)

```
select sName
from (select sailorId, sName
      from Sailor natural join Reserves
      where boatId=101
      intersect
      select sailorId, sName
      from Sailor natural join Reserves
      where boatId=103);
```

Sailors who have
reserved boat 101.

Sailors who have
reserved both boat
101 and boat 103.

Sailors who have
reserved boat 103.

Sailor(sailorId, sName, hkid, rating, age)

Reserves(sailorId, boatId, rDate)

Boat(boatId, bName, color, bType)

EXERCISE 4A) (CONTD)

Use
intersect

Find the sailors who have reserved both boat 101 and boat 103.
Include in the query result tuples attribute sailor name.

☞ (Dustin)

from Sailor natural join Reserves where boatId=101						
sailorId	sName	hkid	rating	age	boatId	rDate
22	Dustin	A298456(6)	7	45	101	10-OCT-25
64	Horatio	F710943(2)	7	35	101	05-SEP-25

Sailors who have reserved boat 101.

select Sailor.sailorId, sName	
sailorId	sName
22	Dustin
64	Horatio

∩ Intersect on sName.

Query
result

sName
Dustin

Why are there
no duplicates
in the result?

from Sailor natural join Reserves where boatId=103						
sailorId	sName	hkid	rating	age	boatId	rDate
22	Dustin	A298456(6)	7	45	103	08-SEP-25
31	Lucy	P930281(5)	10	55	103	06-NOV-25
74	Horatio	T810463(1)	9	35	103	08-SEP-25

Sailors who have reserved boat 103.

select Sailor.sailorId, sName	
sailorId	sName
22	Dustin
31	Lucy
74	Horatio

Sailor(sailorId, sName, hkid, rating, age)

Reserves(sailorId, boatId, rDate)

Boat(boatId, bName, color, bType)

EXERCISE 4A) (CONTD)

Use
intersect

Find the sailors who have reserved both boat 101 and boat 103.
Include in the query result tuples attribute sailor name.

 (Dustin)

```
select sName
from (select sName
      from Sailor natural join Reserves
      where boatId=101
      intersect
      select sName
      from Sailor natural join Reserves
      where boatId=103);
```

Query result: (Dustin) (Horatio)

What happens if we remove sailorId
from the inner select clauses?

Sailors who have
reserved boat 101.

Sailors who have
reserved both boat
101 and boat 103.

Sailors who have
reserved boat 103.

What's the problem?

Sailor(sailorId, sName, hkid, rating, age)

Reserves(sailorId, boatId, rDate)

Boat(boatId, bName, color, bType)

What happens if we remove sailorId from the inner **select** clauses?

EXERCISE 4A) (CONT'D)

Use
intersect

Find the sailors who have reserved both boat 101 and boat 103.
Include in the query result tuples attribute sailor name.

☞ (Dustin)

from Sailor natural join Reserves where boatId=101						
sailorId	sName	hkid	rating	age	boatId	rDate
22	Dustin	A298456(6)	7	45	101	10-OCT-25
64	Horatio	F710943(2)	7	35	101	05-SEP-25

Sailors who have reserved boat 101.

select sName	
sName	
Dustin	
Horatio	

∩ Intersect on sName.

Query
result

sName
Dustin
Horatio

X

from Sailor natural join Reserves where boatId=103						
sailorId	sName	hkid	rating	age	boatId	rDate
22	Dustin	A298456(6)	7	45	103	08-SEP24
31	Lucy	P930281(5)	10	55	103	06-NOV-25
74	Horatio	T810463(1)	9	35	103	08-SEP-25

Sailors who have reserved boat 103.

select sName	
sName	
Dustin	
Lucy	
Horatio	

☞ sName is not
unique!

Sailor(sailorId, sName, hkid, rating, age)

Reserves(sailorId, boatId, rDate)

Boat(boatId, bName, color, bType)

EXERCISE 4B)

Use
JOIN

Find the sailors who have reserved both boat 101 and boat 103.
Include in the query result tuples attribute sailor name.

 (Dustin)

 Hint: You need to use correlation names.

```
select distinct sName
from Sailor S, Reserves R1, Reserves R2
where S.sailorId=R1.sailorId
and R1.boatId=101
and S.sailorId=R2.sailorId
and R2.boatId=103;
```

The same sailor id's must be in
Sailor and in both join results.

Join Sailor and Reserves
on sailorId; keep only
tuples where boatId=101.

Join Sailor and Reserves
on sailorId; keep only
tuples where boatId=103.

Sailor(sailorId, sName, hkid, rating, age)

Reserves(sailorId, boatId, rDate)

Boat(boatId, bName, color, bType)

EXERCISE 4B) (CONTD)

Use
JOIN

Find the sailors who have reserved both boat 101 and boat 103.
Include in the query result tuples attribute sailor name.

👉 (Dustin)

Only sailor id 22 is in Sailor
and in both selection results.

Sailor				
sailorId	sName	hkid	rating	age
10	Emily	P936219(3)	null	16
15	Zoey	K091876(5)	null	17
22	Dustin	A298456(6)	7	45
29	Brutus	D746239(0)	1	33
31	Lucy	P930281(5)	10	55
32	Andy	M028391(4)	8	25
58	Rachael	A819204(3)	10	17
64	Horatio	F710943(2)	7	35
71	Zorba	P910562(4)	1	16
74	Horatio	T810463(1)	9	35
85	Amy	B719037(8)	3	25
95	Bob	E012735(5)	3	63
99	Chris	F710943(2)	10	30

`select * from Reserves where boatId=101;`

R1.boatId=101		
sailorId	boatId	rDate
22	101	10-OCT-25
64	101	05-SEP-25

JOIN sailorId

`select distinct sName.`

Query
result

sName
Dustin

R2.boatId=103		
sailorId	boatId	rDate
22	103	08-SEP-25
31	103	06-NOV-25
74	103	08-SEP-25

`select * from Reserves where boatId=103;`

Sailor(sailorId, sName, hkid, rating, age)

Reserves(sailorId, boatId, rDate)

Boat(boatId, bName, color, bType)

EXERCISE 5

Find sailors who have never reserved a red boat.
Include in the query result tuples attributes sailor id and name.
Order the query result tuples by sailor id ascending.

(10, Emily) (15, Zoey) (29, Brutus) (32, Andy) (58, Rachael) (71, Zorba) (74, Horatio) (85, Amy) (95, Bob)

Project Sailor on sailorId and sName.

sailorId	sName
10	Emily
15	Zoey
22	Dustin
29	Brutus
31	Lucy
32	Andy
58	Rachael
64	Horatio
71	Zorba
74	Horatio
85	Amy
95	Bob
99	Chris

Minus
sailors
who
have
reserved
a red
boat.

```
select sailorId, sName
from Sailor
minus
select sailorId, sName
from Sailor natural join Reserves natural join Boat
where color='red'
order by sailorId;
```

Query result

sailorId	sName
10	Emily
15	Zoey
29	Brutus
32	Andy
58	Rachael
71	Zorba
74	Horatio
85	Amy
95	Bob

Sailors who have never
reserved a red boat.

Project on sailorId and sName.

sailorId	sName
22	Dustin
22	Dustin
31	Lucy
31	Lucy
64	Horatio
99	Chris

Join Sailor and Reserves on sailorId.

sailorId	sName	...	boatId	bName	color	bType
22	Dustin	...	102	Interlake	red	sloop
22	Dustin	...	104	Marine	red	catamaran
31	Lucy	...	102	Interlake	red	sloop
31	Lucy	...	102	Interlake	red	sloop
64	Horatio	...	102	Interlake	red	sloop
99	Chris	...	102	Interlake	red	sloop

Sailors who have
reserved a red boat.



EXERCISE 6

Find any three sailors who are 30 years or older, have reserved a boat and have the highest rating among sailors 30 years or older. Include in the query result tuples attributes sailor name and rating. Order the query result tuples by rating descending.

☞ (Lucy, 10) (Chris, 10) (Horatio, 9)

Is this a
correct
solution?
No! Why?

```
select distinct sName, hkid, rating  
from Sailor natural join Reserves  
where age >= 30  
order by rating desc  
fetch first 3 rows only;
```

Query result: (Lucy, 10) (Chris, 10) (Horatio, 9)

While the query result is correct, *for this database instance*, it will not be correct in general.

What's the problem?

☞ In SQL, null values are the **largest value** in sort order.

EXERCISE 6 (CONT'D)

Find any three sailors who are 30 years or older, have reserved a boat and have the highest rating among sailors 30 years or older. Include in the query result tuples attributes sailor name and rating. Order the query result tuples by rating descending.

☞ (Lucy, 10) (Chris, 10) (Horatio, 9)

```
select distinct sName, hkid, rating
from Sailor natural join Reserves
where age >= 30
order by rating desc nulls last
fetch first 3 rows only;
```



EXERCISE 7

Find boats that have the same name.
Include in the query result tuples attributes boat id and name.
Order the query result tuples by boat id ascending.

👉 (101 Interlake) (102 Interlake)

Is this a
correct
solution?
No! Why?

```
select distinct B1.boatId, B1.bName
from Boat B1, Boat B2
where B1.boatId <> B2.boatId
order by boatId;
```

B1.boatId	B1.bName	B1.color	B1.bType	B2.boatId	B2.bName	B2.color	B2.bType
101	Interlake	blue	dinghy	102	Interlake	red	sloop
101	Interlake	blue	dinghy	103	Clipper	green	dinghy
101	Interlake	blue	dinghy	104	Marine	red	catamaran
101	Interlake	blue	dinghy	105	Serenity	cyan	null
102	Interlake	red	sloop	101	Interlake	blue	dinghy
102	Interlake	red	sloop	103	Clipper	green	dinghy
102	Interlake	red	sloop	104	Marine	red	catamaran
102	Interlake	red	sloop	105	Serenity	Cyan	null
103	Clipper	green	dinghy	101	Interlake	blue	dinghy
103	Clipper	green	dinghy	102	Interlake	red	sloop
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Query result

B1.boatId	B1.bName
101	Interlake
102	Interlake
103	Clipper
104	Marine
105	Serenity

Includes all boats.



EXERCISE 7 (CONTD)

Find boats that have the same name.
Include in the query result tuples attributes boat id and name.
Order the query result tuples by boat id ascending.

👉 (101 Interlake) (102 Interlake)

```
select B1.boatId, B1.bName
from Boat B1, Boat B2
where B1.bName=B2.bName
and B1.boatId<>B2.boatId
order by boatId;
```

B1.boatId	B1.bName	B1.color	B1.bType	B2.boatId	B2.bName	B2.color	B2.bType
101	Interlake	blue	dinghy	102	Interlake	red	sloop
101	Interlake	blue	dinghy	101	Interlake	blue	dinghy
102	Interlake	red	sloop	101	Interlake	blue	dinghy
102	Interlake	red	sloop	102	Interlake	red	sloop
103	Clipper	green	dinghy	103	Clipper	green	dinghy
104	Marine	red	catamaran	104	Marine	red	catamaran
105	Serenity	cyan	null	105	Serenity	cyan	null

Retain only
those tuples
where the boat
names are the
same and the
boat ids are
different.

Join Boat with itself.

EXERCISE 7 (CONTD)

Find the boats that have the same name.
Include in the query result tuples attributes boat id and name.
Order the query result tuples by boat id ascending.

👉 (101 Interlake) (102 Interlake)

```
select B1.boatId, B1.bName
from Boat B1, Boat B2
where B1.bName=B2.bName
and B1.boatId<>B2.boatId
order by boatId;
```

B1.boatId	B1.bName	B1.color	B1.bType	B2.boatId	B2.bName	B2.color	B2.bType
101	Interlake	blue	dinghy	102	Interlake	red	sloop
102	Interlake	red	sloop	101	Interlake	blue	dinghy

Can we use natural join
for this query?

No! Why?

The query result tuples is
empty.

Query result

B1.boatId	B1.bName
101	Interlake
102	Interlake

Project on boatId
and bName.

EXERCISE 7 (CONTD)

Find the boats that have the same name.
Include in the query result tuples attributes boat id and name.
Order the query result tuples by boat id ascending.

👉 (101 Interlake) (102 Interlake)

Is this a
correct
solution?

Yes! but...

~~`select distinct B1.boatId, B1.bName
from Boat B1 join Boat B2 using (bName)
where B1.boatId <> B2.boatId;`~~

👉 Cannot qualify bName in the **select** clause! Why?

bName is used for the natural join and you cannot
qualify on attributes used in a natural join.

`select distinct B1.boatId, bName
from Boat B1 join Boat B2 using (bName)
where B1.boatId <> B2.boatId;`